



National Institute of
Neurological Disorders
and Stroke



Narcolepsy

 English 

What is narcolepsy?

Narcolepsy is a chronic neurological disorder that affects the brain's ability to control [sleep-wake cycles](https://ninds.nih.gov/health-information/public-education/brain-basics/brain-basics-understanding-sleep) (<https://ninds.nih.gov/health-information/public-education/brain-basics/brain-basics-understanding-sleep>). People with narcolepsy may feel rested after waking but then feel very sleepy throughout much of the day. Many individuals with narcolepsy also experience fragmented sleep at night, meaning they can't stay asleep for long periods of time.

In a normal sleep cycle, people enter rapid eye movement (REM) sleep after about 60 to 90 minutes. REM sleep is the sleep stage during which people do the most dreaming. The brain keeps the person's muscles limp during this sleep stage, which prevents them from acting out their dreams.

In people with narcolepsy, the regulation of the sleep and waking cycle is disrupted (called fragmented sleep). Therefore, the boundaries between wakefulness and sleep are less distinct, and elements of sleep and wakefulness can mix. People with narcolepsy frequently enter REM sleep much more quickly, often within 15 minutes of falling asleep. They also can experience muscle weakness or dream activity of REM sleep while they are awake.

Narcolepsy can greatly affect daily activities. If narcolepsy is not diagnosed or treated, it can interfere with emotional well-being, social interactions, and the ability to think clearly, which can affect school, work, and social life.

Symptoms of narcolepsy

Once narcolepsy develops, the person will have it for the rest of their life. Symptoms differ among individuals, and they can improve over time, but will never go away completely. Even when symptoms are severe, they do not result in permanent dysfunction. After episodes end, people rapidly recover the ability to move and speak.

The most typical symptoms are:

- **Excessive daytime sleepiness (EDS)**—All individuals with narcolepsy have periods of EDS. It is often the most obvious symptom. Everyone with narcolepsy has some daytime sleepiness, but the severity of sleepiness varies among individuals. EDS causes severe daytime sleepiness that doesn't improve, even after getting enough sleep at night. Sleepiness in narcolepsy is often like a “sleep attack,” where an overwhelming feeling of sleepiness comes on quickly. In between sleep attacks, individuals can have normal levels of alertness, particularly if doing activities that keep their attention.
- **Cataplexy**—This symptom leads to sudden episodes of muscle weakness, often triggered by strong emotions such as laughter, fear, anger, stress, or excitement. Cataplexy may appear weeks or even years after the onset of EDS. Some people may only have one or two attacks in a lifetime, while others may experience many attacks a day. In about 10% of cases of narcolepsy, cataplexy is the first symptom to appear. Sometimes, it can be misdiagnosed as a seizure disorder. Attacks may be mild and involve only brief, minor weakness in a few muscles, such as a slight drooping of the eyelids. The most severe attacks result in a total body collapse during which individuals are unable to move, speak, or keep their eyes open. Even during the most severe episodes, people remain fully conscious, a characteristic that makes cataplexy different from fainting or seizure disorders. The person remains fully conscious, even if unable to speak, during the episode, which usually lasts a few seconds to several minutes and resolves on their own. While scary, the episodes are not dangerous as long as the individual is in a safe place.
- **Sleep paralysis**—This temporary inability to move or speak while falling asleep or waking up usually lasts only a few seconds or minutes and is similar to the reductions of voluntary muscle activity brought on by REM sleep. Sleep paralysis resembles cataplexy, except that it occurs at the edges of sleep. Very vivid dreamlike and sometimes frightening visual, auditory, or tactile hallucinations can accompany sleep paralysis and usually occur when people are falling asleep or waking up.
- **Fragmented nighttime sleep and insomnia**—While individuals with narcolepsy are very sleepy during the day, they usually also experience difficulties staying asleep at night. They may wake up several times each night for 10–20 minutes, which can worsen daytime sleepiness. Sleep may be disrupted by insomnia, vivid dreaming, [sleep apnea](https://ninds.nih.gov/archived/health-information/disorders/sleep-apnea) (<https://ninds.nih.gov/archived/health-information/disorders/sleep-apnea>), acting out dreams, and periodic leg movements.
- **REM sleep behavior disorder (RBD)**—Usually, when people dream, the body's muscles become temporarily paralyzed. This keeps people from physically acting out their dreams. However, people with RBD don't experience this paralysis. As a result, they might punch, kick, shout, or grab while asleep, disturbing their sleep and the sleep of people around them. RBD can be a problem on its own, or it could be a sign of another neurological disorder like narcolepsy.

Types of narcolepsy

There are two major types of narcolepsy:

1. **Type 1 narcolepsy** (previously known as narcolepsy with cataplexy)—Type 1 narcolepsy is diagnosed in two ways. One way is detecting low levels of hypocretin (also known as orexin), a natural chemical that helps you stay awake and controls REM sleep. The other way is when a person has cataplexy and shows excessive daytime sleepiness during a special nap test.
2. **Type 2 narcolepsy** (previously known as narcolepsy without cataplexy)—People with this condition experience excessive daytime sleepiness but usually do not have cataplexy. They usually also have less severe symptoms and normal hypocretin levels.

A condition known as **secondary narcolepsy** can result from a brain injury to the hypothalamus, a region deep in the brain that helps regulate sleep. In addition to experiencing the typical symptoms of narcolepsy, individuals with secondary narcolepsy may also have other severe neurological problems and sleep for long periods (more than 10 hours) each night.

Who is more likely to get narcolepsy?

Narcolepsy affects men and women equally. Symptoms usually begin between ages 7 and 25 but can start at any age. People with narcolepsy are often misdiagnosed with other conditions like psychiatric disorders, so it can take years to get the correct diagnosis. Nearly all people with narcolepsy type 1 have extremely low levels of hypocretin. Although the causes of narcolepsy are not completely understood, current research suggests that narcolepsy may be the result of a combination of factors working together to cause a lack of hypocretin.

These factors include:

- **Autoimmune disorders**—Autoimmune disorders occur when the body's immune system turns against itself and mistakenly attacks healthy cells or tissue. When cataplexy is present, the cause is most often the loss of brain cells that produce hypocretin. Although the reason for this cell loss is unknown, it appears to be linked to abnormalities in the immune system. Researchers believe that in individuals with narcolepsy, the body's immune system selectively attacks the hypocretin-containing brain cells because of a combination of genetic and environmental factors.
- **Family history**—Most cases of narcolepsy are sporadic, meaning the disorder occurs in individuals with no known family history. However, clusters in families sometimes occur—up to 10% of individuals diagnosed with Type 1 narcolepsy report having a close relative with similar symptoms.
- **Brain injuries**—Rarely, narcolepsy results from traumatic injury to parts of the brain that regulate wakefulness and REM sleep or from tumors and other diseases in the same regions.

Groups of neurons in several parts of the brain interact to control sleep, and many genes control the activity of these neurons. In the past few decades, scientists have made considerable progress in understanding narcolepsy and identifying genes strongly associated with the disorder. The loss of hypocretin neurons in the hypothalamus is the main cause of type 1 narcolepsy. These neurons help stabilize sleep and wake states.

The human leukocyte antigen (HLA) system of genes plays an important role in regulating the immune system. This gene family provides instructions for making a group of related proteins called the HLA complex, which helps the immune system distinguish between good proteins from an individual's own body and bad proteins made by foreign invaders like viruses and bacteria.

One of the genes in this family is HLA-DQB1. A variation in this gene, called HLA-DQB1*06:02, increases the chance of developing narcolepsy, particularly Type 1 narcolepsy. HLA-DQB1*06:02 and other HLA gene variations may increase susceptibility to an immune attack on hypocretin neurons, causing these cells to die. Most people with narcolepsy have this gene variation and may also have specific versions of closely related HLA genes.

Narcolepsy follows a seasonal pattern and is more likely to develop in the spring and early summer after the winter season, a time when people are more likely to get sick. By studying people soon after they develop the disorder, scientists have discovered that individuals with narcolepsy have high levels of certain antibodies, indicating an immune response to a recent bacterial infection (such as strep throat). Also, the H1N1 influenza epidemic in 2009 resulted in a large increase in the number of new cases of narcolepsy. Together, this suggests that individuals with the HLA-DQB1*06:02 gene variation are at risk for developing narcolepsy after they are exposed to a specific trigger, like certain infections that trick the immune system into attacking the body.

How is narcolepsy diagnosed and treated?

Diagnosing narcolepsy

To diagnose narcolepsy, a doctor will need to perform a clinical exam and take a detailed medical history. Individuals may be asked by their doctor to keep a sleep journal noting the times of sleep and symptoms over a one- to two-week period. A physical exam can rule out or identify other neurological conditions that may be causing the symptoms. It is often necessary to visit a sleep medicine specialist for an accurate diagnosis.

Two specialized tests, which can be performed in a sleep disorders clinic, are required to establish a diagnosis of narcolepsy:

1. **Polysomnogram (PSG or sleep study)**—This overnight test records brain activity, muscle movements, breathing, and eye movements during sleep. It helps determine if REM sleep happens too early in the sleep cycle and checks for other conditions like sleep apnea.
2. **Multiple sleep latency test (MSLT)**—This test measures how quickly a person falls asleep and whether they enter REM sleep.

Sometimes, measuring the level of hypocretin in the fluid surrounding the brain and spinal cord can help with diagnosis. To perform this test, a doctor will take a sample of the cerebrospinal fluid using a lumbar puncture (also called a spinal tap) and measure the level of hypocretin-1.

Treating narcolepsy

Although there is no cure for narcolepsy, some of the symptoms can be managed with a combination of medications and lifestyle changes. Treatments target a person's symptoms, rather than the underlying disease. Treatment varies widely by person, and it often takes a long time to find the right combination of treatments.

Research has shown that people with narcolepsy are at a higher risk for heart disease and other serious heart problems. Their heart health should be monitored regularly by a doctor and taken into consideration when considering medications or other therapies.

Medications

Medications are important for managing symptoms, and many people with narcolepsy take multiple medications to

help. Treatments affect each person differently. Therefore, finding effective treatment with medications that produce the greatest benefits with the fewest problems can take time.

- **Wake promoting agents**—Wake promoting agents (like modafinil, armodafinil, and solriamfetol), also known as central nervous system stimulants, are typically the first choice for treatment, because they are less addictive and have fewer side effects than previously used stimulants. For most people, these drugs are generally effective at reducing daytime drowsiness and improving alertness.
- **Amphetamine-like stimulants**—In cases where modafinil is not effective, doctors may prescribe amphetamine-like stimulants (such as methylphenidate) to alleviate EDS. However, these medications must be carefully monitored for serious side effects.
- **Antidepressants**—Two classes of antidepressant drugs have proven effective in controlling cataplexy in many individuals: tricyclics (including imipramine, desipramine, clomipramine, and protriptyline) and selective serotonin and noradrenergic reuptake inhibitors (including venlafaxine, fluoxetine, and atomoxetine).
- **Sodium oxybate**—This drug (also known as gamma hydroxybutyrate or GHB) has been approved by the U.S. Food and Drug Administration (FDA) to treat cataplexy and excessive daytime sleepiness in individuals with narcolepsy. Due to safety concerns associated with its use, the distribution of sodium oxybate is tightly restricted. A similar drug, called mixed salt oxybate (or low sodium oxybate) contains a combination of salts, including calcium, magnesium, and potassium, in addition to sodium. This formulation has significantly less sodium compared to sodium oxybate.
- **Histamine 3 receptor antagonist/inverse agonist**—Pitolisant is the only non-scheduled product for treating excessive daytime sleepiness and cataplexy in adults with narcolepsy. It has also been approved to treat excessive daytime sleepiness in children 6 years of age and older. Pitolisant, which has been commercially available in the U.S. since 2019, is thought to increase histamine levels in the brain.

Lifestyle changes

People with narcolepsy should consider lifestyle changes in addition to medications to best manage symptoms. Consider the following tips:

1. **Take short naps.** Many individuals take short, regularly scheduled naps at times when they tend to feel sleepiest.
2. **Maintain a regular sleep schedule.** Going to bed and waking up at the same time every day, even on the weekends, can help people sleep better.
3. **Avoid caffeine or alcohol before bed.** Individuals should avoid alcohol and caffeine for several hours before bedtime.
4. **Avoid smoking,** especially at night.
5. **Cognitive behavioral therapy (CBT).** CBT, a form of talk therapy, can help people with narcolepsy sleep better, feel less sleepy during the day, and improve their overall health.
6. **Exercise.** Exercising for at least 20 minutes most days at least four or five hours before bedtime improves sleep quality. Regular exercise has been shown to reduce excessive daytime sleepiness in people with narcolepsy.
7. **Eat healthy and avoid large, heavy meals late in the day.** Because of increased risk for heart problems, people with narcolepsy should follow a heart-healthy diet. There is no specific diet for narcolepsy, but emerging research suggests that following a ketogenic diet (high in healthy fats and low in carbohydrates) may help reduce EDS in people with narcolepsy type 1. Eating very close to bedtime can make it harder to sleep.
8. **Relax before bed.** Relaxing activities such as a warm bath before bedtime can help promote sleepiness. Try to keep the sleep space cool and comfortable.
9. **Take safety precautions.** Taking steps to stay safe, especially when driving, is important for people with narcolepsy. Avoiding driving when sleepy is one way to stay safe. Additionally, the Americans with Disabilities Act (ADA) requires employers to allow adults with narcolepsy to ask for work adjustments, like flexible schedules for naps or doing challenging tasks when they are most awake. Kids and teens with narcolepsy can also work with their schools to adjust schedules, take medicine during the day, and use other strategies to manage their condition.

Support groups can help people with narcolepsy and their families develop connections and share stories and strategies with others living with the same condition.

What are the latest updates on narcolepsy?

The mission of NINDS (<https://ninds.nih.gov.com/glossary?title=NINDS>) is to seek fundamental knowledge about the brain and nervous system and to use that knowledge to reduce the burden of neurological disease for all people. NINDS, a component of the National Institutes of Health (NIH (<https://ninds.nih.gov.com/glossary?title=NIH>)), along with several other NIH institutes and centers, supports research on narcolepsy and other sleep disorders through grants to medical institutions across the country.

For example, the National Heart, Lung, and Blood Institute (NHLBI (<https://ninds.nih.gov.com/glossary?title=NHLBI>)) manages the [National Center on Sleep Disorders Research \(NCSDR\)](http://www.nhlbi.nih.gov/about/ncsdr) (<http://www.nhlbi.nih.gov/about/ncsdr>), which coordinates federal government sleep research activities, promotes research training programs, and educates the public and healthcare professionals about sleep disorders.

Genetics and biochemicals

NINDS-sponsored researchers are conducting studies to understand the wide range of genetic factors—both HLA genes and non-HLA genes—and environmental factors that may cause narcolepsy.

Other investigators are using animal models to understand how hypocretin and other biochemicals, such as glutamate, may play a key role in regulating sleep and wakefulness. Biochemicals include proteins, sugars, and fats which help the body function. They are important for digestion, energy production, and cell function.

A greater understanding of the complex genetic and biochemical bases of narcolepsy may help support the development of new therapies to control symptoms and may one day lead to a cure.

Immune system

Abnormalities in the immune system may play an important role in the development of narcolepsy. NINDS (<https://ninds.nih.gov.com/glossary?title=NINDS>) scientists have demonstrated the presence of unusual immune system activity in people with narcolepsy. Further, strep throat and certain varieties of influenza are now thought to trigger narcolepsy in some at-risk individuals. Other NINDS researchers are also working to understand why the immune system destroys hypocretin neurons in narcolepsy.

Sleep biology

NINDS (<https://ninds.nih.gov.com/glossary?title=NINDS>) continues to support investigations into the basic biology of sleep, such as examining the brain mechanisms involved in REM sleep and other sleep behaviors. Since networks of neurons in the brain control sleep and circadian rhythms, NINDS researchers are also examining how brain circuits function in the body and contribute to sleep disorders like narcolepsy. A deeper understanding of sleep biology will help scientists better understand narcolepsy and other sleep disorders.

Orexin-based therapies

Researchers are investigating compounds that help people stay awake to hopefully one day develop new treatment options for narcolepsy that reduce undesired side effects and potential for misuse. Researchers are investigating how new medications, like orexin-based therapies, may help manage narcolepsy. Orexin-based therapies are medical treatments that focus on the orexin system in the brain, which helps control how awake, alert, and hungry we feel. Sleepiness and cataplexy in narcolepsy can happen because the brain doesn't have enough orexin. Research teams are studying how treatments that add more orexin or help it work better can help people stay awake and manage their narcolepsy symptoms.



Learn About Clinical Trials

Clinical trials are studies that allow us to learn more about disorders and improve care. They can help connect patients with new and upcoming treatment options.

Search Clinical Trials <https://clinicaltrials.gov/ct2/home>

How can I or my loved one help improve care for people with narcolepsy?

Consider participating in a clinical trial, so clinicians and scientists can learn more about narcolepsy and related disorders. Current trials are being done to understand immune pathways and how medications, like clarithromycin, reduce sleepiness for narcolepsy.

Clinical research with human participants helps researchers learn more about a disorder and perhaps find better ways to detect, treat, or prevent disease safely.

All types of participants are needed—those who are healthy or may have an illness or disease—of all different ages, sexes, races, and ethnicities to ensure that study results apply to as many people as possible and that treatments will be safe and effective for everyone who will use them.

For information about participating in clinical research visit [NIH Clinical Research Trials and You](http://www.nih.gov/health/clinicaltrials) (<http://www.nih.gov/health/clinicaltrials>). Learn about clinical trials currently looking for people with narcolepsy at [Clinicaltrials.gov](http://clinicaltrials.gov/) (<http://clinicaltrials.gov/>).

The [NeuroBioBank](https://neurobiobank.nih.gov/) (<https://neurobiobank.nih.gov/>) is a resource for scientists studying neurological disorders, including

narcolepsy. Donating tissue from individuals with narcolepsy can help researchers better understand the disorder. The NeuroBioBank connects brain and tissue repositories, researchers, NIH (<https://ninds.nih.gov.com/glossary?title=NIH>) staff, disease advocacy groups, and people interested in donating.

Additionally, NINDS (<https://ninds.nih.gov.com/glossary?title=NINDS>) supports genetic and immunological research in narcolepsy at the [Stanford University Center for Narcolepsy](https://med.stanford.edu/narcolepsy.html) (<https://med.stanford.edu/narcolepsy.html>). Blood samples from individuals with narcolepsy can be sent by mail.

NINDS acknowledges [Dr. Roneil Malkani](https://www.feinberg.northwestern.edu/faculty-profiles/az/profile.html?xid=23422) (<https://www.feinberg.northwestern.edu/faculty-profiles/az/profile.html?xid=23422>) and the [American Academy of Neurology](https://www.aan.com/) (<https://www.aan.com/>) for their contributions to this piece.

Where can I find more information about narcolepsy?

Information may be available from the following organizations:

[Narcolepsy Network](https://narcolepsynetwork.org/) (<https://narcolepsynetwork.org/>)
401-667-2523
info@narcolepsynetwork.org
(<mailto:info@narcolepsynetwork.org>)

[National Heart, Lung, and Blood Institute \(NHLBI\)](https://www.nhlbi.nih.gov/) (<https://www.nhlbi.nih.gov/>)
301-592-8573
nhlbiinfo@nhlbi.nih.gov (<mailto:nhlbiinfo@nhlbi.nih.gov>)

[National Library of Medicine](https://www.nlm.nih.gov/) (<https://www.nlm.nih.gov/>)
301-594-5983
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(<mailto:pubmedcentral@ncbi.nlm.nih.gov>)

[National Sleep Foundation](https://www.sleepfoundation.org/) (<https://www.sleepfoundation.org/>)
703-243-1697
nsf@sleepfoundation.org (<mailto:nsf@sleepfoundation.org>)



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